

Sphere

A three-dimensional, ball-shaped, perfectly round object, where each point on its surface is the same distance from its center.

Spread

The spread of a set of data is how the data is distributed over a range.

Square

A quadrilateral in which all the angles are the same (90°) and every side is the same length.

Square root

A number that, multiplied by itself, produces a given number, shown as $\sqrt{\quad}$, e.g. $\sqrt{4} = 2$.

Squared number

The result of multiplying a number by itself, e.g. $4^2 = 4 \times 4 = 16$.

Standard deviation

A measure of spread that shows the amount of deviation from the mean. If the standard deviation is low the data is close to the mean, if it is high, it is widely spread.

Standard form

A number (usually very large or very small) written as a positive or negative number between 1 and 9 multiplied by a power of 10, e.g. 0.02 is 2×10^{-2} .

Statistics

The collection, presentation, and interpretation of data.

Stem-and-leaf diagram

A graph showing the shape of ordered data. Numbers are split in two digits and separated by a line. The first digits form the stem (written once) and the second digits form leaves (written many times in rows).

Substitution

Putting something in place of something else, e.g. using a constant number in place of a variable.

Subtraction/subtract

Taking a number away from another number. It is represented by the symbol $-$.

Sum

The total, or the number calculated when two numbers are added together.

Supplementary angle

Two angles that add up to 180° .

Symmetry/symmetrical

A shape or object is symmetrical if it looks the same after a reflection or a rotation.

Table

Information displayed in rows and columns.

Take-home pay

Take-home pay is the amount of earnings left after tax has been paid.

Tangent

1. A straight line that touches a curve at one point.
2. In trigonometry, tangent is the ratio of the side opposite to a given angle with the side adjacent to the given angle, in a right-angled triangle.

Tax

Money that is paid to the government, either as part of what a person buys, or as a part of their income.

Terms

Individual numbers in a sequence or series, or individual parts of an expression, e.g. in $7a^2 + 4xy - 5$ the terms are $7a^2$, $4xy$, and 5 .

Tessellation

A pattern of shapes covering a surface without leaving any gaps.

Theoretical probability

The likelihood of an outcome based on mathematical ideas rather than experiments.

Three-dimensional

Objects that have length, width, and height. Three-dimensions is often written as 3D.

Transformation

A change of position, size, or orientation. Reflections, rotations, enlargements, and translations are all transformations.

Translation

Movement of an object without it being rotated.

Trapezoid

A quadrilateral with a pair of parallel sides that can be of different lengths.

Triangle

A two-dimensional shape with 3 sides and 3 angles.

Trigonometry

The study of triangles and the ratios of their sides and angles.

Two-dimensional

A flat figure that has length and width. Two-dimensions is often written as 2D.

Unit

1. The standard amount in measuring, e.g. cm, kg, and seconds.
2. Another name for one.

Unknown angle

An angle which is not specified, and for which the number of degrees need to be determined.

Variable

A quantity that can vary or change and is usually indicated by a letter.

Vector

A quantity that has both size and direction, e.g. velocity and force are vectors.

Velocity

The speed and direction in which something is moving, measured in metres per second m/s.

Vertex (plural: vertices)

The corner or point at which surfaces or lines meet.

Vertical

At right-angles to the horizon. A vertical line goes between up and down directions.

Volume

The amount of space within a three-dimensional object. Volume is measured in units cubed, e.g. cm^3 .

Wage

The amount of money paid to a person in exchange for work.

Whole number

Counting numbers that do not have any fractional parts and are greater than or equal to 0, e.g. 1, 7, 46, 108.

Whole turn

A rotation of 360° , so that an object faces the same direction it started from.

Width

The sideways length, measuring between opposite sides. Width is the same as breadth.

X-axis

The horizontal axis of a graph, which determines the x-coordinate.

X-intercept

The value at which a line crosses the x-axis on a graph.

Y-axis

The vertical axis of a graph, which determines the y-coordinate.